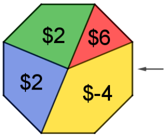
MATH1231 Mathematics 1B – Algebra

Section 12 - Discrete Random Variables

Lecturer: Aidan Sims (aidan.sims@unsw.edu.au)
Slides: David Angell, Laure Helme-Guizon and others

Example 1. In this spinner game, playing consists of choosing one of the colours. You win \$6 if red is drawn, you win \$2 if green or blue is drawn, and you loose \$4 if yellow is drawn. Let X be the amount the player wins when they play once. (The amount is negative if the player loses.)

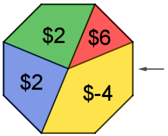


a) What the probability that someone (playing once) will loose money?

A random variable is an assignment of a numerical value to each possible outcome of a random phenomenon.

b) Is X as defined in this spinner game a random variable?

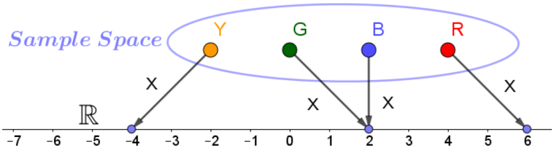
Example 1. In this spinner game, playing consists of choosing one of the colours. You win \$6 if red is drawn, you win \$2 if green or blue is drawn, and you loose \$4 if yellow is drawn. Let X be the amount the player wins when they play once. (The amount is negative if the player loses.)



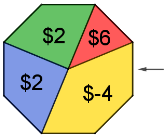
a) What the probability that someone (playing once) will loose money?

A random variable is an assignment of a numerical value to each possible outcome of a random phenomenon.

b) Is X as defined in this spinner game a random variable?



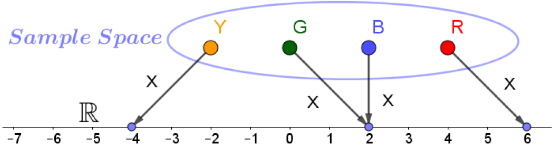
Example 1. In this spinner game, playing consists of choosing one of the colours. You win \$6 if red is drawn, you win \$2 if green or blue is drawn, and you loose \$4 if yellow is drawn. Let X be the amount the player wins when they play once. (The amount is negative if the player loses.)



a) What the probability that someone (playing once) will loose money?

A random variable is an assignment of a numerical value to each possible outcome of a random phenomenon.

b) Is X as defined in this spinner game a random variable?



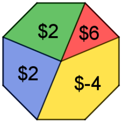
So ... a random variable is not random?

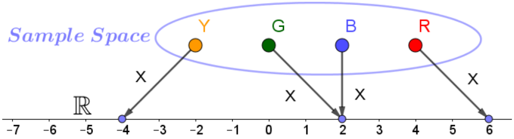
Definition (**random variable**)

In Statistics, a **random variable** is an assignment of a numerical value to each possible outcome of a random phenomenon.
In other words, a random variable on a sample space S is a function $X : S \to \mathbb{R}$.
We write
$$\mathbb{P}(\mathbf{X} = \mathbf{x}) \overset{def}{=} \mathbb{P}(\{ s \in S \mid X(s) = x \}) ,$$
with similar meanings for $\mathbb{P}(\mathbf{X} \leqslant \mathbf{x})$, $\mathbb{P}(\mathbf{X} \in \mathbf{A})$ and so on.

This definition perhaps does not sound very helpful. As a rough but possibly more accessible idea, you can think of a random variable as being a quantity which we can measure or observe, and which need not be the same every time we measure it.

Our guiding example. Let X be the amount the player wins when they play once.





Let X be the (signed) amount the player wins when they play once.
c) Find $\mathbb{P}(X = 2)$.

....continued

The examples on this slide illustrate that (1) a random variable is an assignment of a numerical value to each possible outcome of a random phenomenon (2) random variables can be used to define events and (3) that many different random variables can be attached to the same random experiment and we pick or define one that corresponds to our needs.

Example 2. Consider the sample space consisting of all possible sequences of three coin tosses. There are numerous random variables which might be of interest.

1. Let $X(s)$ be 1 if s includes a head, 0 if not; then, for instance,

The examples on this slide illustrate that (1) a random variable is an assignment of a numerical value to each possible outcome of a random phenomenon (2) random variables can be used to define events and (3) that many different random variables can be attached to the same random experiment and we pick or define one that corresponds to our needs.

Example 2. Consider the sample space consisting of all possible sequences of three coin tosses. There are numerous random variables which might be of interest.

1. Let $X(s)$ be 1 if s includes a head, 0 if not; then, for instance,
$$X(HTT) = 1 \quad \text{and} \quad \mathbb{P}(X = 0) = \mathbb{P}(\{\, TTT \,\}) = \frac{1}{8} \,.$$

2. Let $Y(s)$ be the number of tails in s; then

The examples on this slide illustrate that (1) a random variable is an assignment of a numerical value to each possible outcome of a random phenomenon (2) random variables can be used to define events and (3) that many different random variables can be attached to the same random experiment and we pick or define one that corresponds to our needs.

Example 2. Consider the sample space consisting of all possible sequences of three coin tosses. There are numerous random variables which might be of interest.

1. Let $X(s)$ be 1 if s includes a head, 0 if not; then, for instance,
$$X(HTT) = 1 \quad \text{and} \quad \mathbb{P}(X = 0) = \mathbb{P}(\{\, TTT \,\}) = \frac{1}{8}\ .$$

2. Let $Y(s)$ be the number of tails in s; then
$$Y(HTT) = 2 \quad \text{and} \quad \mathbb{P}(Y \geqslant 2) = \mathbb{P}(\{\, TTT,\ TTH,\ THT,\ HTT \,\}) = \frac{1}{2}\ .$$

3. Let $Z(s)$ be the number of heads after the last tail, or 3 if there is no tail; then

The examples on this slide illustrate that (1) a random variable is an assignment of a numerical value to each possible outcome of a random phenomenon (2) random variables can be used to define events and (3) that many different random variables can be attached to the same random experiment and we pick or define one that corresponds to our needs.

Example 2. Consider the sample space consisting of all possible sequences of three coin tosses. There are numerous random variables which might be of interest.

1. Let $X(s)$ be 1 if s includes a head, 0 if not; then, for instance,
$$X(HTT) = 1 \quad \text{and} \quad \mathbb{P}(X = 0) = \mathbb{P}(\{\, TTT \,\}) = \frac{1}{8}\ .$$
2. Let $Y(s)$ be the number of tails in s; then
$$Y(HTT) = 2 \quad \text{and} \quad \mathbb{P}(Y \geqslant 2) = \mathbb{P}(\{\, TTT,\ TTH,\ THT,\ HTT \,\}) = \frac{1}{2}\ .$$
3. Let $Z(s)$ be the number of heads after the last tail, or 3 if there is no tail; then
$$Z(THT) = 0\ , \quad Z(HHH) = 3 \quad \text{and} \quad \mathbb{P}(Z = 1) = \mathbb{P}(\{\, HTH,\ TTH \,\}) = \frac{1}{4}\ .$$

....continued

Example 3. Let S be the Australian population;

- for any person s in S, let $X(s)$ be the age of s, in years, on their last birthday before 1 June 2015. Then, for example, $X(\text{Nicole Kidman}) = 56$; and figures from the 'people 100 years old' example can be written as

$$\mathbb{P}(X \geqslant 100) = 0.00019\ , \quad \mathbb{P}(X = 0) = 0.013\ , \quad \mathbb{P}(X < 18) = 0.22\ .$$

Example 3. Let S be the Australian population;

- for any person s in S, let $X(s)$ be the age of s, in years, on their last birthday before 1 June 2015. Then, for example, $X(\text{Nicole Kidman}) = 56$; and figures from the 'people 100 years old' example can be written as

$$\mathbb{P}(X \geqslant 100) = 0.00019 , \quad \mathbb{P}(X = 0) = 0.013 , \quad \mathbb{P}(X < 18) = 0.22 .$$

- We could also define $Y(s)$ to be the exact age of s, in years, at 00:00 on 1 June 2015; note that Y is not the same as X.

Example 3. Let S be the Australian population;

- for any person s in S, let $X(s)$ be the age of s, in years, on their last birthday before 1 June 2015. Then, for example, $X(\text{Nicole Kidman}) = 56$; and figures from the 'people 100 years old' example can be written as

$$\mathbb{P}(X \geqslant 100) = 0.00019 , \quad \mathbb{P}(X = 0) = 0.013 , \quad \mathbb{P}(X < 18) = 0.22 .$$

- We could also define $Y(s)$ to be the exact age of s, in years, at 00:00 on 1 June 2015; note that Y is not the same as X.

Notations. *Please, please understand and respect the notations. It makes things so much clearer for you and for us!*

- We commonly use CAPITAL letters for random variables and lowercase letters for the values taken by random variables.

- Make sure you clearly understand that something like $\mathbb{P}(X = x)$ is not an equation – it is just a number (specifically, a probability).

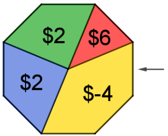
Definition (cumulative distribution function)

Let X be a random variable. The **cumulative distribution function** of X is denoted by F_X, or if there is no ambiguity just by F, and is the function

$$F_X : \mathbb{R} \to \mathbb{R} \quad \text{where} \quad F_X(x) \overset{\text{def}}{=} \mathbb{P}(X \leqslant x) \, .$$

Example 4, the spinner game.

Let X be the amount the player wins when they play once. (The amount is negative if the player loses.)



a) $F_X(-5) = \mathbb{P}(\, \dots\dots\dots \,) = \dots$

b) $F_X(0) =$

c) $F_X(2) =$

d) $F_X(5) =$

e) $F_X(6) =$

f) $F_X(17) =$

g) $F_X(6) - F_X(2) = \mathbb{P}(\, \dots\dots\dots\dots\dots \,) = \dots$

Previous slide: Let X be a random variable. The **cumulative distribution function** of X is denoted by F_X, or if there is no ambiguity just by F, and is the function

$$F_X : \mathbb{R} \to \mathbb{R} \quad \text{where} \quad F_X(x) \stackrel{\text{def}}{=} \mathbb{P}\left(X \leqslant x\right) .$$

Fact (Properties of cumulative distribution function)

If F is the cumulative distribution function of a random variable X, then

- F is non-decreasing, meaning, if $a \leqslant b$ then $F(a) \leqslant F(b)$;
- if $a \leqslant b$ then $\mathbb{P}\left(a < X \leqslant b\right) = F(b) - F(a)$;
- $\lim\limits_{x \to -\infty} F(x) = 0$ and $\lim\limits_{x \to \infty} F(x) = 1$.

Investigation via example 5. Consider two random variables defined on the sample space T of all times from $8:00$am to $9:00$am today:

- $X(t)$ is the number of passengers on my bus at time t;
- $Y(t)$ is the distance (in kilometres) of my bus from the UNSW Anzac Pde stop at time t.

What makes these random variable fundamentally different?

Investigation via example 5. Consider two random variables defined on the sample space T of all times from $8:00$am to $9:00$am today:

- $X(t)$ is the number of passengers on my bus at time t;
- $Y(t)$ is the distance (in kilometres) of my bus from the UNSW Anzac Pde stop at time t.

What makes these random variable fundamentally different?

The values taken by X are the non–negative integers $0, 1, 2, \ldots$, continuing indefinitely in principle (though not in practice – even on a Sydney bus!). The values taken by Y, on the other hand, are not restricted to be integers but include arbitrary real numbers (though again there will be, in practice, bounds on these values).

Fundamentally different types of random variables

Investigation via example 5. Consider two random variables defined on the sample space T of all times from $8:00$am to $9:00$am today:

- $X(t)$ is the number of passengers on my bus at time t;
- $Y(t)$ is the distance (in kilometres) of my bus from the UNSW Anzac Pde stop at time t.

What makes these random variable fundamentally different?

The values taken by X are the non–negative integers $0, 1, 2, \ldots$, continuing indefinitely in principle (though not in practice – even on a Sydney bus!). The values taken by Y, on the other hand, are not restricted to be integers but include arbitrary real numbers (though again there will be, in practice, bounds on these values).

An important consequence of this difference is that while it makes sense to ask about the probability that X has a specific value, it is much less clear what this means in the case of Y. For example, $\mathbb{P}(X = 5)$ can be regarded as the total amount of time (in hours) during which there are 5 passengers on the bus. But $\mathbb{P}(Y = 5)$ should mean the probability that the bus is *exactly* 5km from the stop. Since it is impossible to measure distance exactly, and in any case the bus would only be at this distance for an instant, the only sensible value for this probability is zero. But then we have the problem of how to interpret a zero probability for an event which is not impossible.

Investigation via example 5. Consider two random variables defined on the sample space T of all times from $8:00$am to $9:00$am today:

- $X(t)$ is the number of passengers on my bus at time t;
- $Y(t)$ is the distance (in kilometres) of my bus from the UNSW Anzac Pde stop at time t.

What makes these random variable fundamentally different?

The values taken by X are the non–negative integers $0, 1, 2, \ldots$, continuing indefinitely in principle (though not in practice – even on a Sydney bus!). The values taken by Y, on the other hand, are not restricted to be integers but include arbitrary real numbers (though again there will be, in practice, bounds on these values).

An important consequence of this difference is that while it makes sense to ask about the probability that X has a specific value, it is much less clear what this means in the case of Y. For example, $\mathbb{P}(X = 5)$ can be regarded as the total amount of time (in hours) during which there are 5 passengers on the bus. But $\mathbb{P}(Y = 5)$ should mean the probability that the bus is *exactly* 5km from the stop. Since it is impossible to measure distance exactly, and in any case the bus would only be at this distance for an instant, the only sensible value for this probability is zero. But then we have the problem of how to interpret a zero probability for an event which is not impossible.

Roughly speaking, random variables like X are referred to as discrete, and those like Y are continuous.

Definition (**discrete** random variable)

A random variable $X : S \to \mathbb{R}$ is said to be **discrete** if its range (or image),

$$\{ X(s) \mid s \in S \} ,$$

is countable (i.e. it can be written as a list, finite or infinite).

Example 6.

1. The random variable X on the previous page is discrete; Y is not.
2. Let S be the set of all sequences of 3 coin tosses, and let X be the random variable giving the number of heads in a sequence.

Definition (**discrete** random variable)

A random variable $X : S \to \mathbb{R}$ is said to be **discrete** if its range (or image),

$$\{ X(s) \mid s \in S \} ,$$

is countable (i.e. it can be written as a list, finite or infinite).

Example 6.

1. The random variable X on the previous page is discrete; Y is not.
2. Let S be the set of all sequences of 3 coin tosses, and let X be the random variable giving the number of heads in a sequence. Then X is discrete because it can only take the four values $0, 1, 2, 3$.
3. A coin is tossed repeatedly until the first head occurs; let X be the number of tosses.

Definition (**discrete** random variable)

A random variable $X : S \to \mathbb{R}$ is said to be **discrete** if its range (or image),

$$\{ X(s) \mid s \in S \} ,$$

is countable (i.e. it can be written as a list, finite or infinite).

Example 6.

1. The random variable X on the previous page is discrete; Y is not.
2. Let S be the set of all sequences of 3 coin tosses, and let X be the random variable giving the number of heads in a sequence. Then X is discrete because it can only take the four values $0, 1, 2, 3$.
3. A coin is tossed repeatedly until the first head occurs; let X be the number of tosses. Then the range of X, that is, the set of values of X, is $\{ 1, 2, 3, \dots \}$ and so X is discrete.
4. In the spinner game, X, the amount the player wins when they play once is ☐ *discrete* ☐ *continuous*.
5. Find an example of a continuous random variable: ...

Definition (probability distribution)

The **probability distribution** of a random variable X is some description of the probabilities of all events associated with X.

If X is a discrete random variable, we often denote its range as $\{x_1, x_2, x_3, \dots\}$ and the associated probabilities as $p_k = P(X = x_k)$ for each positive integer k.

Definition (probability distribution)

The **probability distribution** of a random variable X is some description of the probabilities of all events associated with X.

If X is a discrete random variable, we often denote its range as $\{x_1, x_2, x_3, \dots\}$ and the associated probabilities as $p_k = P(X = x_k)$ for each positive integer k.

We can describe the probability distribution of a discrete random variable X by listing all the outcomes $\{x_1, x_2, x_3, \dots\}$ in a sequence or table together with the corresponding probabilities.

A probability distribution is valid if and only if:

- $p_k \geqslant 0$ for all positive integers k, and
- $\displaystyle\sum_k p_k = 1$.

Definition (probability distribution)

The **probability distribution** of a random variable X is some description of the probabilities of all events associated with X.

If X is a discrete random variable, we often denote its range as $\{x_1, x_2, x_3, \dots\}$ and the associated probabilities as $p_k = P(X = x_k)$ for each positive integer k.

We can describe the probability distribution of a discrete random variable X by listing all the outcomes $\{x_1, x_2, x_3, \dots\}$ in a sequence or table together with the corresponding probabilities.

A probability distribution is valid if and only if:

- $p_k \geqslant 0$ for all positive integers k, and
- $\sum_k p_k = 1$.

In fact, it is impossible for an uncountable collection of positive numbers to have sum equal to 1: this is why it is important to make the distinction between random variables with countable range, where each element can be assigned a positive probability, and those with uncountable range, where this cannot be done.

Example 7. The random variable X is given by the sum of the rolls of two 6-sided dice. Find the probability distribution of X, and find $P(X \geqslant 10)$.

Probability distribution

Example 7. The random variable X is given by the sum of the rolls of two 6-sided dice. Find the probability distribution of X, and find $P(X \geqslant 10)$.

Solution. First we can use an addition table to list all 36 possible roll totals:

+	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

Probability distribution

Example 7. The random variable X is given by the sum of the rolls of two 6-sided dice. Find the probability distribution of X, and find $P(X \geqslant 10)$.

Solution. First we can use an addition table to list all 36 possible roll totals:

+	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

Since each entry in the above table represents an equally likely outcome, the probability distribution can be summarised by the following table:

x_k	2	3	4	5	6	7	8	9	10	11	12
p_k	$\frac{1}{36}$		$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$		$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

Probability distribution

Example 7. The random variable X is given by the sum of the rolls of two 6-sided dice. Find the probability distribution of X, and find $P(X \geqslant 10)$.

Solution. First we can use an addition table to list all 36 possible roll totals:

+	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

Since each entry in the above table represents an equally likely outcome, the probability distribution can be summarised by the following table:

x_k	2	3	4	5	6	7	8	9	10	11	12
p_k	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

So the (tail) probability that the total is at least 10 is given by

$$P(X \geqslant 10) =$$

Probability distribution

Example 7. The random variable X is given by the sum of the rolls of two 6-sided dice. Find the probability distribution of X, and find $P(X \geqslant 10)$.

Solution. First we can use an addition table to list all 36 possible roll totals:

$+$	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

Since each entry in the above table represents an equally likely outcome, the probability distribution can be summarised by the following table:

x_k	2	3	4	5	6	7	8	9	10	11	12
p_k	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

So the (tail) probability that the total is at least 10 is given by

$$P(X \geqslant 10) = P(X = 10) + P(X = 11) + P(X = 12) = \frac{3}{36} + \frac{2}{36} + \frac{1}{36} = \frac{1}{6}.$$

Unknown probability distribution

Example 8. A 6-sided die is weighted so that rolling a value of k is k times more likely than rolling a value of 1. If the random variable X is given by the result of the roll, find the probability distribution of X, and find $P(X < 4)$.

Unknown probability distribution

Example 8. A 6-sided die is weighted so that rolling a value of k is k times more likely than rolling a value of 1. If the random variable X is given by the result of the roll, find the probability distribution of X, and find $P(X < 4)$.

Solution. Let the probability of rolling a 1 be p, that is, $P(X = 1) = p$. Then an intermediary probability distribution of X can be summarised by the following table:

x_k	1	2	3	4	5	6
p_k						

Example 8. A 6-sided die is weighted so that rolling a value of k is k times more likely than rolling a value of 1. If the random variable X is given by the result of the roll, find the probability distribution of X, and find $P(X < 4)$.

Solution. Let the probability of rolling a 1 be p, that is, $P(X = 1) = p$. Then an intermediary probability distribution of X can be summarised by the following table:

x_k	1	2	3	4	5	6
p_k	p					

Unknown probability distribution

Example 8. A 6-sided die is weighted so that rolling a value of k is k times more likely than rolling a value of 1. If the random variable X is given by the result of the roll, find the probability distribution of X, and find $P(X<4)$.

Solution. Let the probability of rolling a 1 be p, that is, $P(X=1)=p$. Then an intermediary probability distribution of X can be summarised by the following table:

x_k	1	2	3	4	5	6
p_k	p	$2p$				

Unknown probability distribution

Example 8. A 6-sided die is weighted so that rolling a value of k is k times more likely than rolling a value of 1. If the random variable X is given by the result of the roll, find the probability distribution of X, and find $P(X < 4)$.

Solution. Let the probability of rolling a 1 be p, that is, $P(X = 1) = p$. Then an intermediary probability distribution of X can be summarised by the following table:

x_k	1	2	3	4	5	6
p_k	p	$2p$	$3p$	$4p$	$5p$	$6p$

Unknown probability distribution

Example 8. A 6-sided die is weighted so that rolling a value of k is k times more likely than rolling a value of 1. If the random variable X is given by the result of the roll, find the probability distribution of X, and find $P(X<4)$.

Solution. Let the probability of rolling a 1 be p, that is, $P(X=1)=p$. Then an intermediary probability distribution of X can be summarised by the following table:

x_k	1	2	3	4	5	6
p_k	p	$2p$	$3p$	$4p$	$5p$	$6p$

Recalling that the sum of all probabilities must be 1, we have $p+2p+3p+4p+5p+6p=1$, so $p=\frac{1}{21}$. So the exact probability distribution of X can be summarised by the following table:

x_k	1	2	3	4	5	6
p_k	$\frac{1}{21}$	$\frac{2}{21}$	$\frac{3}{21}$	$\frac{4}{21}$	$\frac{5}{21}$	$\frac{6}{21}$

Example 8. A 6-sided die is weighted so that rolling a value of k is k times more likely than rolling a value of 1. If the random variable X is given by the result of the roll, find the probability distribution of X, and find $P(X<4)$.

Solution. Let the probability of rolling a 1 be p, that is, $P(X=1)=p$. Then an intermediary probability distribution of X can be summarised by the following table:

x_k	1	2	3	4	5	6
p_k	p	$2p$	$3p$	$4p$	$5p$	$6p$

Recalling that the sum of all probabilities must be 1, we have $p+2p+3p+4p+5p+6p=1$, so $p=\frac{1}{21}$. So the exact probability distribution of X can be summarised by the following table:

x_k	1	2	3	4	5	6
p_k	$\frac{1}{21}$	$\frac{2}{21}$	$\frac{3}{21}$	$\frac{4}{21}$	$\frac{5}{21}$	$\frac{6}{21}$

So the probability that the total is less than 4 is given by

Unknown probability distribution

Example 8. A 6-sided die is weighted so that rolling a value of k is k times more likely than rolling a value of 1. If the random variable X is given by the result of the roll, find the probability distribution of X, and find $P(X<4)$.

Solution. Let the probability of rolling a 1 be p, that is, $P(X=1) = p$. Then an intermediary probability distribution of X can be summarised by the following table:

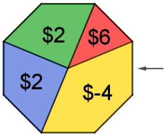
x_k	1	2	3	4	5	6
p_k	p	$2p$	$3p$	$4p$	$5p$	$6p$

Recalling that the sum of all probabilities must be 1, we have $p + 2p + 3p + 4p + 5p + 6p = 1$, so $p = \frac{1}{21}$. So the exact probability distribution of X can be summarised by the following table:

x_k	1	2	3	4	5	6
p_k	$\frac{1}{21}$	$\frac{2}{21}$	$\frac{3}{21}$	$\frac{4}{21}$	$\frac{5}{21}$	$\frac{6}{21}$

So the probability that the total is less than 4 is given by
$P(X<4) = P(X=3) + P(X=2) + P(X=1) = \frac{3}{21} + \frac{2}{21} + \frac{1}{21} = \frac{2}{7}$.

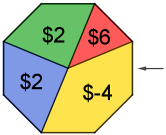
Example 9. In this spinner game, playing consists of choosing one of the colours. You win \$6 if red is drawn, you win \$2 if green or blue is drawn, and you loose \$4 if yellow is drawn. Let X be the amount the player wins when they play once. (The amount is negative if the player loses.)



How much can a player expect to win if they play this game 800 times?

Investigation leading to a definition

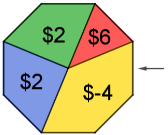
Example 9. In this spinner game, playing consists of choosing one of the colours. You win \$6 if red is drawn, you win \$2 if green or blue is drawn, and you loose \$4 if yellow is drawn. Let X be the amount the player wins when they play once. (The amount is negative if the player loses.)



How much can a player expect to win if they play this game 800 times?

Investigation leading to a definition

Example 9. In this spinner game, playing consists of choosing one of the colours. You win \$6 if red is drawn, you win \$2 if green or blue is drawn, and you loose \$4 if yellow is drawn. Let X be the amount the player wins when they play once. (The amount is negative if the player loses.)



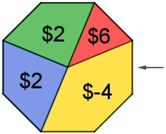
How much can a player expect to win if they play this game 800 times?

This motivates the following definition.

Definition (expected value of a discrete random variable)

The **expected value** or **mean** of a discrete random variable X, written $\mathbb{E}(\mathbf{X})$ (or μ_X or just μ), is given by

$$\mathbb{E}(X) = \sum_k x_k p_k .$$



Example 9. In this spinner game, playing consists of choosing one of the colours. You win \$6 if red is drawn, you win \$2 if green or blue is drawn, and you loose \$4 if yellow is drawn. Let X be the amount the player wins when they play once. (The amount is negative if the player loses.)

How much can a player expect to win if they play this game 800 times?

This motivates the following definition.

Definition (expected value of a discrete random variable)

The **expected value** or **mean** of a discrete random variable X, written $\mathbb{E}(\mathbf{X})$ (or μ_X or just μ), is given by

$$\mathbb{E}(X) = \sum_k x_k p_k .$$

In other words, the expected value is a weighted average of all possible outcomes, where the weights are the probability of each outcome.

Expected value or mean

Previous slide: The expected value or mean of X is $\mathbb{E}(X) = \mu = \sum_k x_k p_k$.

That thing is called the mean? I have heard that word before. Is it related?

It is. If the range of X is finite (of size n, say) and all outcomes are equally likely, we have

$$\mathbb{E}(X) = \frac{1}{n}(x_1 + x_2 + \cdots + x_n) = \frac{x_1 + x_2 + \cdots + x_n}{n},$$

which is just our good old arithmetic mean of $\{x_1, x_2, ..., x_n\}$.

Expected value or mean

Previous slide: The expected value or mean of X is $\mathbb{E}(X) = \mu = \sum_k x_k p_k$.

That thing is called the mean? I have heard that word before. Is it related?

It is. If the range of X is finite (of size n, say) and all outcomes are equally likely, we have

$$\mathbb{E}(X) = \frac{1}{n}(x_1 + x_2 + \cdots + x_n) = \frac{x_1 + x_2 + \cdots + x_n}{n},$$

which is just our good old arithmetic mean of $\{x_1, x_2, ..., x_n\}$.

Believe it or not, the ***expected*** *is not a value you necessarily* ***expect*** *to get.*

Example 9. Suppose X is the random variable which outputs the result of a single roll of a 6-sided die. Then the expected value of X is

$$\mathbb{E}(X) = 1 \times \frac{1}{6} + 2 \times \frac{1}{6} + 3 \times \frac{1}{6} + 4 \times \frac{1}{6} + 5 \times \frac{1}{6} + 6 \times \frac{1}{6} = \frac{7}{2}.$$

Expected value or mean

Previous slide: The expected value or mean of X is $\mathbb{E}(X) = \mu = \sum_k x_k p_k$.

That thing is called the mean? I have heard that word before. Is it related?

It is. If the range of X is finite (of size n, say) and all outcomes are equally likely, we have

$$\mathbb{E}(X) = \frac{1}{n}(x_1 + x_2 + \cdots + x_n) = \frac{x_1 + x_2 + \cdots + x_n}{n},$$

which is just our good old arithmetic mean of $\{x_1, x_2, ..., x_n\}$.

Believe it or not, the ***expected*** *is not a value you necessarily* ***expect*** *to get.*

Example 9. Suppose X is the random variable which outputs the result of a single roll of a 6-sided die. Then the expected value of X is

$$\mathbb{E}(X) = 1 \times \frac{1}{6} + 2 \times \frac{1}{6} + 3 \times \frac{1}{6} + 4 \times \frac{1}{6} + 5 \times \frac{1}{6} + 6 \times \frac{1}{6} = \frac{7}{2}.$$

Notice that $\frac{7}{2} = 3.5$ is not a possible output when you roll a die, but does represent the value we would expect to approach if we averaged the results of the experiment repeated many times.

Example 10. A fair coin is toss until a head is obtained. Let X be the number of tosses to get the first head.

a) What is the probability of getting head for the first time on the 3rd toss?

b) Find $p_5 = \mathbb{P}(X = 5)$.

c) Find the probability distribution of X.

d) Find $\mathbb{E}(X)$.

Hint: $\frac{k}{2^k} = \frac{k+1}{2^{k-1}} - \frac{k+2}{2^k}$

Example 10. A fair coin is toss until a head is obtained. Let X be the number of tosses to get the first head.

a) What is the probability of getting head for the first time on the 3rd toss?

b) Find $p_5 = \mathbb{P}(X = 5)$.

c) Find the probability distribution of X.
The number of tosses to get the first head is

$$p_k = \mathbb{P}(X = k) = \frac{1}{2^k} \quad \text{for} \quad k = 1, 2, 3, \dots \ .$$

d) Find $\mathbb{E}(X)$. *Hint:* $\frac{k}{2^k} = \frac{k+1}{2^{k-1}} - \frac{k+2}{2^k}$

Example 10. A fair coin is toss until a head is obtained. Let X be the number of tosses to get the first head.

a) What is the probability of getting head for the first time on the 3rd toss?

b) Find $p_5 = \mathbb{P}(X = 5)$.

c) Find the probability distribution of X.
The number of tosses to get the first head is

$$p_k = \mathbb{P}(X = k) = \frac{1}{2^k} \quad \text{for} \quad k = 1, 2, 3, \dots \; .$$

d) Find $\mathbb{E}(X)$. *Hint:* $\frac{k}{2^k} = \frac{k+1}{2^{k-1}} - \frac{k+2}{2^k}$

- Realising it is a '*telescoping series*' or a '*domino series*' gives a way of evaluating this infinite series.

$$\begin{aligned}
\sum_{k=1}^{n} \frac{k}{2^k} &= \sum_{k=1}^{n} \left(\frac{k+1}{2^{k-1}} - \frac{k+2}{2^k} \right) \\
&= \left(\frac{2}{2^0} + \frac{3}{2^1} + \frac{4}{2^2} + \cdots + \frac{n+1}{2^{n-1}} \right) - \left(\frac{3}{2^1} + \frac{4}{2^2} + \cdots + \frac{n+1}{2^{n-1}} + \frac{n+2}{2^n} \right) \\
&= 2 - \frac{n+2}{2^n}
\end{aligned}$$

and so $E(X) = \lim_{n\to\infty} \sum_{k=1}^{n} \frac{k}{2^k} = \lim_{n\to\infty} \left(2 - \frac{n+2}{2^n} \right) = 2$.

- Alternatively, once Calculus catches up, we can differentiate term by term the power series $\sum_{k=1}^{\infty} x^k = \frac{x}{1-x}$ for $|x| < 1$ and then substituting $x = \frac{1}{2}$.

Example 11.

- A die is loaded so that the numbers show up with unequal probabilities:

x_k	1	2	3	4	5	6
p_k	0.1	0.1	0.2	0.1	0.2	0.3

Can we get $\mathbb{E}(f(X))$ from $\mathbb{E}(X)$ and f? Maybe $f(\mathbb{E}(X))$?

Example 12.

- A die is loaded so that the numbers show up with unequal probabilities:

x_k	1	2	3	4	5	6
p_k	0.1	0.1	0.2	0.1	0.2	0.3

Let X be the random variable giving the outcome of a roll of this die. Then the average number obtained by rolling the die is

$$\mathbb{E}(X) = 1(0.1) + 2(0.1) + 3(0.2) + 4(0.1) + 5(0.2) + 6(0.3) = 4.1\ .$$

Can we get $\mathbb{E}(f(X))$ from $\mathbb{E}(X)$ and f? Maybe $f(\mathbb{E}(X))$?

Example 13.

- A die is loaded so that the numbers show up with unequal probabilities:

x_k	1	2	3	4	5	6
p_k	0.1	0.1	0.2	0.1	0.2	0.3

Let X be the random variable giving the outcome of a roll of this die. Then the average number obtained by rolling the die is

$$\mathbb{E}(X) = 1(0.1) + 2(0.1) + 3(0.2) + 4(0.1) + 5(0.2) + 6(0.3) = 4.1 \ .$$

- Now consider a second die, loaded in the same way, but with the numbers $x = 1, 2, 3, 4, 5, 6$ on the faces replaced by $y = 60/x = 60, 30, 20, 15, 12, 10$ respectively. The random variable Y giving the outcome of a roll of the second die has probability distribution

x_k	60	30	10	14	12	10
p_k	0.1	0.1	0.2	0.1	0.2	0.3

and so the average number obtained is

$$\mathbb{E}(Y) = 60(0.1) + 30(0.1) + 20(0.2) + 15(0.1) + 12(0.2) + 10(0.3) = 19.9 \ .$$

Can we get $\mathbb{E}(f(X))$ from $\mathbb{E}(X)$ and f? Maybe $f(\mathbb{E}(X))$?

Example 14.

- A die is loaded so that the numbers show up with unequal probabilities:

x_k	1	2	3	4	5	6
p_k	0.1	0.1	0.2	0.1	0.2	0.3

Let X be the random variable giving the outcome of a roll of this die. Then the average number obtained by rolling the die is

$$\mathbb{E}(X) = 1(0.1) + 2(0.1) + 3(0.2) + 4(0.1) + 5(0.2) + 6(0.3) = 4.1 \;.$$

- Now consider a second die, loaded in the same way, but with the numbers $x = 1, 2, 3, 4, 5, 6$ on the faces replaced by $y = 60/x = 60, 30, 20, 15, 12, 10$ respectively. The random variable Y giving the outcome of a roll of the second die has probability distribution

x_k	60	30	10	14	12	10
p_k	0.1	0.1	0.2	0.1	0.2	0.3

and so the average number obtained is

$$\mathbb{E}(Y) = 60(0.1) + 30(0.1) + 20(0.2) + 15(0.1) + 12(0.2) + 10(0.3) = 19.9 \;.$$

Note that $Y = \dfrac{60}{X}$ but $\mathbb{E}(Y) \neq \dfrac{60}{\mathbb{E}(X)}$.

Can we get $\mathbb{E}(f(X))$ from $\mathbb{E}(X)$ and f? Maybe $f(\mathbb{E}(X))$?

Example 15.

- A die is loaded so that the numbers show up with unequal probabilities:

x_k	1	2	3	4	5	6
p_k	0.1	0.1	0.2	0.1	0.2	0.3

Let X be the random variable giving the outcome of a roll of this die. Then the average number obtained by rolling the die is

$$\mathbb{E}(X) = 1(0.1) + 2(0.1) + 3(0.2) + 4(0.1) + 5(0.2) + 6(0.3) = 4.1 \ .$$

- Now consider a second die, loaded in the same way, but with the numbers $x = 1, 2, 3, 4, 5, 6$ on the faces replaced by $y = 60/x = 60, 30, 20, 15, 12, 10$ respectively. The random variable Y giving the outcome of a roll of the second die has probability distribution

x_k	60	30	10	14	12	10
p_k	0.1	0.1	0.2	0.1	0.2	0.3

and so the average number obtained is

$$\mathbb{E}(Y) = 60(0.1) + 30(0.1) + 20(0.2) + 15(0.1) + 12(0.2) + 10(0.3) = 19.9 \ .$$

Note that $Y = \dfrac{60}{X}$ but $\mathbb{E}(Y) \neq \dfrac{60}{\mathbb{E}(X)}$.
And indeed, in most cases, $\mathbb{E}(f(X)) \neq f(\mathbb{E}(X))$.

So how do we calculate $\mathbb{E}(f(X))$ since our previous attempt failed?

Note that if X is a random variable on a sample space S and g is a function from $\mathbb{R}$ to $\mathbb{R}$, then $Y = g(X)$ is a function from S to $\mathbb{R}$ and hence, by definition, is a random variable on S. In the case $g(x) = 60/x$ we have the situation of the previous example; in effect, we found the expected value of Y from the calculation

$$\mathbb{E}(Y) = E\left(\frac{60}{X}\right) = \sum_{\text{all } k} \frac{60}{x_k}\, p_k = \sum_{\text{all } k} g(x_k) p_k \ .$$

So how do we calculate $\mathbb{E}(f(X))$ since our previous attempt failed?

Note that if X is a random variable on a sample space S and g is a function from $\mathbb{R}$ to $\mathbb{R}$, then $Y = g(X)$ is a function from S to $\mathbb{R}$ and hence, by definition, is a random variable on S. In the case $g(x) = 60/x$ we have the situation of the previous example; in effect, we found the expected value of Y from the calculation

$$\mathbb{E}(Y) = E\left(\frac{60}{X}\right) = \sum_{\text{all } k} \frac{60}{x_k} p_k = \sum_{\text{all } k} g(x_k) p_k \ .$$

The same formula works for any function!

Theorem (Expected value of a function of a discrete random variable)

Let X be a discrete random variable taking values x_k with probabilities p_k. Let g be a function from $\mathbb{R}$ to $\mathbb{R}$. Then the random variable $Y = g(X)$ has mean

$$\mathbb{E}(Y) = \sum_{\text{all } k} g(x_k) p_k \ .$$

Proof. Omitted – see the printed notes if you wish.

Variance

The mean gives us a sense of the "typical location" of the outcome of a random process.

Variance

The mean gives us a sense of the "typical location" of the outcome of a random process. As well as this, it is also often important to know how the outcomes are ***spread around the mean****.*

The mean gives us a sense of the "typical location" of the outcome of a random process. As well as this, it is also often important to know how the outcomes are ***spread around the mean****.*

For example, if you were planning what to take on a holiday, the following alternative probability distributions of daily maximum temperatures at your destination, both of which have mean 25, would make a considerable difference to your packing.

t	24	25	26	27
p_t	0·13	0·76	0·09	0·02

t	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
p_t	0·01	0·02	0·02	0·03	0·04	0·05	0·06	0·08	0·08	0·09	0·09	0·11	0·22	0·08	0·02

The mean gives us a sense of the "typical location" of the outcome of a random process. As well as this, it is also often important to know how the outcomes are ***spread around the mean****.*

For example, if you were planning what to take on a holiday, the following alternative probability distributions of daily maximum temperatures at your destination, both of which have mean 25, would make a considerable difference to your packing.

t	24	25	26	27
p_t	0·13	0·76	0·09	0·02

t	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
p_t	0·01	0·02	0·02	0·03	0·04	0·05	0·06	0·08	0·08	0·09	0·09	0·11	0·22	0·08	0·02

We can measure the spread by calculating the average distance of X from the mean. If we write $\mu = \mathbb{E}(X)$, then some attempts at doing this are

- $E(X - \mu)$, which is completely useless (why?);
- $E(|X - \mu|)$, which is possible but messy;
- or. . .

Definition (variance and standard deviation of a random variable)

Let X be a random variable with mean μ. The **variance** of X is

$$\mathbb{V}\mathrm{ar}(\mathbf{X}) \stackrel{def}{=} \mathbb{E}((X-\mu)^2)\ .$$

Definition (variance and standard deviation of a random variable)

Let X be a random variable with mean μ. The **variance** of X is

$$\mathbb{V}\mathrm{ar}(\mathbf{X}) \stackrel{def}{=} \mathbb{E}((X-\mu)^2) \ .$$

The **standard deviation** of X, denoted $\mathbf{SD(X)}$, or $\sigma_{\mathbf{X}}$, is the (positive) square root of the variance, i.e.

$$\mathbf{SD(X)} = \sigma_{\mathbf{X}} \stackrel{def}{=} \sqrt{\mathbb{V}\mathrm{ar}(X)} \ .$$

Definition (variance and standard deviation of a random variable)

Let X be a random variable with mean μ. The **variance** of X is

$$\mathbb{V}\mathrm{ar}(\mathbf{X}) \stackrel{def}{=} \mathbb{E}((X-\mu)^2) \; .$$

The **standard deviation** of X, denoted $\mathbf{SD(X)}$, or $\sigma_{\mathbf{X}}$, is the (positive) square root of the variance, i.e.

$$\mathbf{SD(X)} = \sigma_{\mathbf{X}} \stackrel{def}{=} \sqrt{\mathbb{V}\mathrm{ar}(X)} \; .$$

Comments.

- The variance is another instance of the expected value of a function of a random variable.

Definition (variance and standard deviation of a random variable)

Let X be a random variable with mean μ. The **variance** of X is

$$\mathbb{V}\mathrm{ar}(\mathbf{X}) \stackrel{def}{=} \mathbb{E}((X-\mu)^2) \ .$$

The **standard deviation** of X, denoted $\mathbf{SD(X)}$, or $\sigma_{\mathbf{X}}$, is the (positive) square root of the variance, i.e.

$$\mathbf{SD(X)} = \sigma_{\mathbf{X}} \stackrel{def}{=} \sqrt{\mathbb{V}\mathrm{ar}(X)} \ .$$

Comments.

- The variance is another instance of the expected value of a function of a random variable.
- Units in a variance are the squares of the units in the original measurements of a random variable (for example, if X is a length, then $\mathbb{V}\mathrm{ar}(X)$ will "dimensionally" be an area); therefore the variance and the original measurements cannot be directly compared. This is one of the reasons why the standard deviation is important.

Comparing standard deviations gives a way to compare spreads.

Example 16. If T is the temperature distribution from the first table on page 11, we have $\mathbb{E}(T) = 25$ and

$$\begin{aligned}\mathbb{V}\mathrm{ar}(T) &= \mathbb{E}((T-25)^2)\\ &= \sum_{\text{all } t}(t-25)^2 p_t\\ &= (24-25)^2(0.13) + (25-25)^2(0.76) + (26-25)^2(0.09) + (27-25)^2(0.02)\\ &= 0.30\ ,\\ \mathrm{SD}(T) &= 0.55\ .\end{aligned}$$

If, on the other hand, T is given by the second table, then a similar calculation gives

$$\begin{aligned}\mathbb{V}\mathrm{ar}(T) &= 81(0.01) + 64(0.02) + 49(0.02) + \cdots + 25(0.02)\\ &= 11.18\ ,\\ \mathrm{SD}(T) &= \sqrt{11.18} = 3.34\ .\end{aligned}$$

The greater spread of outcomes in the second example is indicated by the greater standard deviation.

Comparing standard deviations gives a way to compare spreads.

Example 16. If T is the temperature distribution from the first table on page 11, we have $\mathbb{E}(T) = 25$ and

$$\begin{aligned}
\mathbb{V}\mathrm{ar}(T) & \\
&= \sum_{\text{all } t} (t-25)^2 p_t \\
&= (24-25)^2(0.13) + (25-25)^2(0.76) + (26-25)^2(0.09) + (27-25)^2(0.02) \\
&= 0.30\ , \\
\mathrm{SD}(T) &= 0.55\ .
\end{aligned}$$

If, on the other hand, T is given by the second table, then a similar calculation gives

$$\begin{aligned}
\mathbb{V}\mathrm{ar}(T) &= 81(0.01) + 64(0.02) + 49(0.02) + \cdots + 25(0.02) \\
&= 11.18\ , \\
\mathrm{SD}(T) &= \sqrt{11.18} = 3.34\ .
\end{aligned}$$

The greater spread of outcomes in the second example is indicated by the greater standard deviation.

Example 16. If T is the temperature distribution from the first table on page 11, we have $\mathbb{E}(T) = 25$ and

$$
\begin{aligned}
\mathbb{V}\mathrm{ar}(T) & \\
&= (24-25)^2(0.13) + (25-25)^2(0.76) + (26-25)^2(0.09) + (27-25)^2(0.02) \\
&= 0.30\ , \\
\mathrm{SD}(T) &= 0.55\ .
\end{aligned}
$$

If, on the other hand, T is given by the second table, then a similar calculation gives

$$
\begin{aligned}
\mathbb{V}\mathrm{ar}(T) &= 81(0.01) + 64(0.02) + 49(0.02) + \cdots + 25(0.02) \\
&= 11.18\ , \\
\mathrm{SD}(T) &= \sqrt{11.18} = 3.34\ .
\end{aligned}
$$

The greater spread of outcomes in the second example is indicated by the greater standard deviation.

Comparing standard deviations gives a way to compare spreads.

Example 16. If T is the temperature distribution from the first table on page 11, we have $\mathbb{E}(T) = 25$ and

$$\begin{aligned}
\mathbb{V}\mathrm{ar}(T) &= \mathbb{E}((T-25)^2) \\
&= \sum_{\text{all } t} (t-25)^2 p_t \\
&= (24-25)^2(0.13) + (25-25)^2(0.76) + (26-25)^2(0.09) + (27-25)^2(0.02) \\
&= 0.30\ , \\
\mathrm{SD}(T) &= 0.55\ .
\end{aligned}$$

Example 16. If T is the temperature distribution from the first table on page 11, we have $\mathbb{E}(T) = 25$ and

$$\begin{aligned}
\mathbb{V}\mathrm{ar}(T) &= \mathbb{E}((T-25)^2) \\
&= \sum_{\text{all } t} (t-25)^2 p_t \\
&= (24-25)^2(0.13) + (25-25)^2(0.76) + (26-25)^2(0.09) + (27-25)^2(0.02) \\
&= 0.30 \ , \\
\mathrm{SD}(T) &= 0.55 \ .
\end{aligned}$$

If, on the other hand, T is given by the second table, then a similar calculation gives

$$\begin{aligned}
\mathbb{V}\mathrm{ar}(T) &= 81(0.01) + 64(0.02) + 49(0.02) + \cdots + 25(0.02) \\
&= 11.18 \ , \\
\mathrm{SD}(T) &= \sqrt{11.18} = 3.34 \ .
\end{aligned}$$

Comparing standard deviations gives a way to compare spreads.

Example 16. If T is the temperature distribution from the first table on page 11, we have $\mathbb{E}(T) = 25$ and

$$\begin{aligned}
\mathbb{V}\mathrm{ar}(T) &= \mathbb{E}((T-25)^2) \\
&= \sum_{\text{all } t} (t-25)^2 p_t \\
&= (24-25)^2(0.13) + (25-25)^2(0.76) + (26-25)^2(0.09) + (27-25)^2(0.02) \\
&= 0.30\ , \\
\mathrm{SD}(T) &= 0.55\ .
\end{aligned}$$

If, on the other hand, T is given by the second table, then a similar calculation gives

$$\begin{aligned}
\mathbb{V}\mathrm{ar}(T) &= 81(0.01) + 64(0.02) + 49(0.02) + \cdots + 25(0.02) \\
&= 11.18\ , \\
\mathrm{SD}(T) &= \sqrt{11.18} = 3.34\ .
\end{aligned}$$

The greater spread of outcomes in the second example is indicated by the greater standard deviation.

There is an alternative formula for the variance, which is sometimes easier to calculate.

There is an alternative formula for the variance, which is sometimes easier to calculate.

Theorem (alternative formula for the variance)

If X is a random variable, then

$$\mathbb{V}\mathrm{ar}(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2 \ .$$

There is an alternative formula for the variance, which is sometimes easier to calculate.

Theorem (alternative formula for the variance)

If X is a random variable, then

$$\mathbb{V}\mathrm{ar}(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2 .$$

Mnemonic: "The mean of the square minus the square of the mean".

Proof for a discrete random variable.

There is an alternative formula for the variance, which is sometimes easier to calculate.

Theorem (alternative formula for the variance)

If X is a random variable, then

$$\mathbb{V}\mathrm{ar}(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2 .$$

Mnemonic: "The mean of the square minus the square of the mean".

Proof for a discrete random variable. Write $\mu = \mathbb{E}(X)$. Then

$$\begin{aligned}
\mathbb{V}\mathrm{ar}(X) &= \sum_{\text{all } k} (x_k - \mu)^2 p_k \\
&= \sum_{\text{all } k} (x_k^2 - 2\mu x_k + \mu^2) p_k \\
&= \left(\sum_{\text{all } k} x_k^2 p_k\right) - 2\mu \left(\sum_{\text{all } k} x_k p_k\right) + \mu^2 \left(\sum_{\text{all } k} p_k\right) \\
&= \mathbb{E}(X^2) - 2\mu\mathbb{E}(X) + \mu^2 \\
&= \mathbb{E}(X^2) - (\mathbb{E}(X))^2 .
\end{aligned}$$

There is no clear connection between means and variances of random variables related by a function such as $Y = 60/X$. But for random variables which are related by "scaling and shifting", there are simple relationships.

There is no clear connection between means and variances of random variables related by a function such as $Y = 60/X$. But for random variables which are related by "scaling and shifting", there are simple relationships.

Theorem (scaling and shifting a random variable)

Let X be a random variable, and let a and b be real constants. Then the mean, variance and standard deviation of the random variable $aX + b$ are given by

$$\mathbb{E}(aX + b) = a\mathbb{E}(X) + b$$

$$\mathbb{V}\mathrm{ar}(aX + b) = a^2\mathbb{V}\mathrm{ar}(X)$$

$$\mathrm{SD}(aX + b) = |a|\,\mathrm{SD}(X)\ .$$

Proof for a discrete random variable.

There is no clear connection between means and variances of random variables related by a function such as $Y = 60/X$. But for random variables which are related by "scaling and shifting", there is simple relationships.

Theorem (scaling and shifting a random variable)

Let X be a random variable, and let a and b be real constants. Then the mean, variance and standard deviation of the random variable $aX + b$ are given by

$$\mathbb{E}(aX + b) = a\mathbb{E}(X) + b$$
$$\mathbb{V}\mathrm{ar}(aX + b) = a^2\mathbb{V}\mathrm{ar}(X)$$
$$\mathrm{SD}(aX + b) = |a|\,\mathrm{SD}(X)\ .$$

Proof for a discrete random variable. We have

$$\mathbb{E}(aX + b) = \sum_{\text{all } k}(ax_k + b)p_k = a\Big(\sum_{\text{all } k} x_k p_k\Big) + b\Big(\sum_{\text{all } k} p_k\Big) = a\mathbb{E}(X) + b\ .$$

Applying this result to the random variable $(X - \mathbb{E}(X))^2$ and the constant a^2, we have

$$\begin{aligned}\mathbb{V}\mathrm{ar}(aX + b) &= \mathbb{E}\big(((aX + b) - (a\mathbb{E}(X) + b))^2\big)\\ &= \mathbb{E}\big(a^2(X - \mathbb{E}(X))^2\big)\\ &= a^2\mathbb{E}\big((X - \mathbb{E}(X))^2\big)\\ &= a^2\mathbb{V}\mathrm{ar}(X)\ .\end{aligned}$$

If we think carefully about what these results mean, we can see them as just common sense!

For example, let C be a random variable which measures a temperature in degrees Celsius, and let F be the random variable which measures the same temperature in degrees Fahrenheit. Then

$$F = \frac{9}{5}C + 32 \ .$$

If we think carefully about what these results mean, we can see them as just common sense!

For example, let C be a random variable which measures a temperature in degrees Celsius, and let F be the random variable which measures the same temperature in degrees Fahrenheit. Then

$$F = \frac{9}{5}C + 32 \ .$$

• Since the same temperatures are being measured, it makes sense that the mean should be the same actual temperature in each case: we only need to convert it from one scale to the other. That is,

$$\mathbb{E}(F) = \frac{9}{5}\mathbb{E}(C) + 32 \ .$$

If we think carefully about what these results mean, we can see them as just common sense!

For example, let C be a random variable which measures a temperature in degrees Celsius, and let F be the random variable which measures the same temperature in degrees Fahrenheit. Then

$$F = \frac{9}{5}C + 32 \ .$$

• Since the same temperatures are being measured, it makes sense that the mean should be the same actual temperature in each case: we only need to convert it from one scale to the other. That is,

$$\mathbb{E}(F) = \frac{9}{5}\mathbb{E}(C) + 32 \ .$$

• Similarly, the spread of temperatures, as indicated by the standard deviation, will be the same for each case "in reality". Shifting the data will not change the spread about the mean, so the 32 is irrelevant; we just have to write the spread in Fahrenheit units instead of Celsius. That is,

$$\mathrm{SD}(F) = \left|\frac{9}{5}\right| \mathrm{SD}(C) = \frac{9}{5}\mathrm{SD}(C) \ .$$

There are numerous discrete probability distributions which arise frequently in practice and are therefore worth studying; in this course we look at just two of them.

The Binomial Distribution

The Geometric Distribution

Definition (Bernoulli trial)

A **Bernoulli trial** is an experiment with two possible outcomes, often referred to as "success" and "failure". We shall usually write p for the probability of success in a Bernoulli trial and $q = 1 - p$ for the probability of failure.

There are numerous discrete probability distributions which arise frequently in practice and are therefore worth studying; in this course we look at just two of them.

The Binomial Distribution

The Geometric Distribution

Definition (Bernoulli trial)

A **Bernoulli trial** is an experiment with two possible outcomes, often referred to as "success" and "failure". We shall usually write p for the probability of success in a Bernoulli trial and $q = 1 - p$ for the probability of failure.

Examples of Bernoulli trials:

a) tossing a coin and regarding "heads" as a success, so $p = \frac{1}{2}$;
b) rolling two dice and regarding a six on either or both as a success, so $p = \frac{11}{36}$;
c) selecting a "1kg" box of cereal from a warehouse store room, weighing it, and recording a success if it really is 1kg or more: p unknown!

Definition (Bernoulli process)

A **Bernoulli process** is an experiment consisting of a sequence of

- *identical* Bernoulli trials
- in which the events S_k of a success on the kth trial are *mutually independent*,
- and the number of trials is *fixed and determined beforehand*.

Definition (Bernoulli process)

A **Bernoulli process** is an experiment consisting of a sequence of

- *identical* Bernoulli trials
- in which the events S_k of a success on the kth trial are *mutually independent*,
- and the number of trials is *fixed and determined beforehand*.

Examples of Bernoulli processes:

a) tossing a coin 10 times;

b) rolling two dice 20 times;

c) selecting a cereal box 30 times, *provided* that we replace the box after weighing it and take care to ensure that all boxes, including those selected already, have the appropriate chance of being selected in the future.

Theorem (Number of successes in a Bernoulli process)

Consider a Bernoulli process consisting of n trials with success probability p. If X is the random variable giving the total **number** of successes in this process, then

$$\mathbb{P}(X = k) = \binom{n}{k} p^k q^{n-k} \quad \text{where} \quad q = 1 - p \quad \text{for} \quad k = 0, 1, 2, \ldots, n \, .$$

Proof.

Theorem (Number of successes in a Bernoulli process)

Consider a Bernoulli process consisting of n trials with success probability p. If X is the random variable giving the total **number** of successes in this process, then

$$\mathbb{P}(X = k) = \binom{n}{k} p^k q^{n-k} \quad \text{where} \quad q = 1 - p \quad \text{for} \quad k = 0, 1, 2, \ldots, n \, .$$

Proof. Since trials are mutually independent, any specific sequence of k successes and $n-k$ failures has probability $p^k q^{n-k}$; and there are $\binom{n}{k}$ such sequences (which correspond to branches with k successes in the probability tree).

Definition (binomial distribution)

The **binomial distribution** with n trials and success probability p is the function

$$B(n, p, k) = \binom{n}{k} p^k (1-p)^{n-k} \quad \text{for} \quad k = 0, 1, 2, \ldots, n \, .$$

If X is a random variable with $\mathbb{P}(X = k) = B(n, p, k)$ we say that X has the binomial distribution with n trials and success probability p, and we write $X \sim B(n, p)$.

Notation. The symbol "$\sim$" is read as "*has the (following) distribution*".

Example 17. A jar contains 20 red jelly beans and 8 black jelly beans. A trial consists of shaking the jar, picking a jelly bean at random, recording its colour and ...

1) ... replacing it in the jar. Let X_1 be the proportion of black jelly beans obtained after 10 trials. Does X_1 have a binomial distribution? _____

2) ... replacing it in the jar. Let X_2 be the number of red jelly beans obtained after 10 trials. Does X_2 have a binomial distribution? _____

3) ... replacing it in the jar. Let X_3 be the number of black jelly beans obtained after 10 trials. Does X_3 have a binomial distribution? _____

4) ... eating it. Let X_4 be the number of red jelly beans obtained after 10 trials. Does X_4 have a binomial distribution? _____

5) ... replacing it in the jar. Let X_5 be the number of black jelly beans obtained before the first red one is obtained. Does X_5 have a binomial distribution? __

Exercise 18. Recall that $B(n, p, k) = \binom{n}{k} p^k q^{n-k}$, where $q = 1 - p$. Prove that the values $B(n, p, k)$ really do form a probability distribution.

Solution.

Exercise 18. Recall that $B(n,p,k) = \binom{n}{k} p^k q^{n-k}$, where $q = 1 - p$.
Prove that the values $B(n,p,k)$ really do form a probability distribution.

Solution. It is clear that $B(n,p,k) \geqslant 0$;
and using the binomial theorem gives

$$\sum_{\text{all } k} B(n,p,k) = \sum_{k=0}^{n} \binom{n}{k} p^k q^{n-k} = (p+q)^n = 1 \,.$$

We can use similar ideas and Calculus to evaluate the mean and variance of the binomial distribution, see course notes page 195.

Theorem (Mean and variance of the binomial distribution)

If $X \sim B(n,p)$, then $\mathbb{E}(X) = np$ and $\mathbb{V}\mathrm{ar}(X) = npq$.

Exercise 18. A jar contains 20 red jelly beans and 8 black jelly beans. A trial consists of shaking the jar, picking a jelly bean at random, recording its colour and replacing it in the jar. Let J be the number of black jelly beans obtained after 10 trials. Find the expected value and the standard deviation of J.

Solution.

Exercise 18. Recall that $B(n,p,k) = \binom{n}{k} p^k q^{n-k}$, where $q = 1 - p$. Prove that the values $B(n,p,k)$ really do form a probability distribution.

Solution. It is clear that $B(n,p,k) \geqslant 0$; and using the binomial theorem gives

$$\sum_{\text{all } k} B(n,p,k) = \sum_{k=0}^{n} \binom{n}{k} p^k q^{n-k} = (p+q)^n = 1 \ .$$

We can use similar ideas and Calculus to evaluate the mean and variance of the binomial distribution, see course notes page 195.

Theorem (Mean and variance of the binomial distribution)

If $X \sim B(n,p)$, then $\mathbb{E}(X) = np$ and $\mathbb{V}\mathrm{ar}(X) = npq$.

Exercise 18. A jar contains 20 red jelly beans and 8 black jelly beans. A trial consists of shaking the jar, picking a jelly bean at random, recording its colour and replacing it in the jar. Let J be the number of black jelly beans obtained after 10 trials. Find the expected value and the standard deviation of J.

Solution.
$\mathbb{E}(J) = 10 \times \frac{2}{7} = \frac{20}{7}$ and $\mathbb{V}\mathrm{ar}(J) = 10 \times \frac{2}{7} \times \frac{5}{7} = \frac{100}{49}$ and $\mathrm{SD}(J) = \frac{10}{7}$.

Exercise 19. [time permitting]

Prove that if $X \sim B(n, p)$, then $\mathbb{E}(X) = np$ and $\mathbb{V}\mathrm{ar}(X) = npq$.

Hint: Express $f(x) = (x + q)^n$ using the binomial theorem, differentiate, multiply by x, and then substitute $x = p$.

Solution.

Exercise 19. [time permitting]
Prove that if $X \sim B(n, p)$, then $\mathbb{E}(X) = np$ and $\mathbb{V}\mathrm{ar}(X) = npq$.

Hint: Express $f(x) = (x+q)^n$ using the binomial theorem, differentiate, multiply by x, and then substitute $x = p$.

Solution. With $q = 1 - p$,

$$f(x) = (x+q)^n = \sum_{k=0}^{n} \binom{n}{k} x^k q^{n-k}$$

$$xf'(x) = nx(x+q)^{n-1} = \sum_{k=0}^{n} k \binom{n}{k} x^k q^{n-k}$$

$$\mathbb{E}(X) = \sum_{k=0}^{n} k \binom{n}{k} p^k q^{n-k} = pf'(p) = np(p+q)^{n-1} = np$$

$$x\frac{d}{dx}(xf'(x)) = nx(x+q)^{n-1} + n(n-1)x^2(x+q)^{n-2} = \sum_{k=0}^{n} k^2 \binom{n}{k} x^k q^{n-k}$$

$$\mathbb{E}(X^2) = \sum_{k=0}^{n} k^2 \binom{n}{k} p^k q^{n-k} = np + n(n-1)p^2$$

$$\mathbb{V}\mathrm{ar}(X) = np + n(n-1)p^2 - (np)^2 = np - np^2 = npq \ .$$

Earlier, we considered the number of tosses of a coin required to obtain the first head. This is an example of another important distribution.

The Geometric Distribution

Investigation / Example 20. As before, we have a jar of 20 red and 8 black jelly beans, and we select jelly beans randomly one at a time. Since we do not like red jelly beans, we will always put them back and keep drawing until we select (and eat) a black jelly bean. Let X be the number of draws it takes to obtain the first black jelly bean.

a) Find the probability that it takes (exactly) 10 draws to successfully obtain a black jelly bean.

b) Find $\mathbb{P}(X = 10)$

c) Find $\mathbb{P}(X = k)$, k is positive integer.

Theorem (**First success in a Bernoulli process**)

Consider a Bernoulli process consisting of infinitely many trials with success probability p. If X is the random variable giving the number of trials required to obtain the first success in this process, then

$$\mathbb{P}(X = k) = q^{k-1}p \quad \text{for} \quad k = 1, 2, 3, \ldots \ .$$

Proof. The first success occurs on trial k if and only if the first $k-1$ trials are failures and the kth is a success.

Definition (**geometric distribution**)

The **geometric distribution** with success probability p is the function

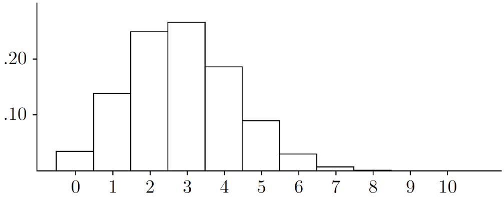
$$G(p, k) = (1-p)^{k-1}p \quad \text{for} \quad k = 1, 2, 3, \ldots \ .$$

If $\mathbb{P}(X = k) = G(p, k)$ then we say X has a geometric distribution with success probability p and we write $X \sim G(p)$.

Note that the sample space for a geometric distribution problem will omit the sequence consisting of infinitely many failures and no successes, so that "$X = \infty$" is impossible. As the omitted sequence has zero probability, this will not cause any difficulty in practice.

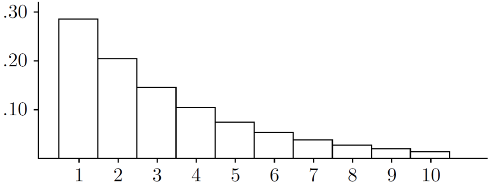
In both example below, we have a jar of 20 red and 8 black jelly beans, and we select jelly beans randomly, one at a time, with replacement.

- Probabilities of getting k black jelly beans in $n = 10$ trials.



Note that the balance point is $\mathbb{E}(X) = np = 10 \times \frac{2}{7} \approx 3$.

- Probabilities that it takes k trials to get a black jelly bean for the first time.



The balance point is $\mathbb{E}(Y)$ again (it always is) ... but what is $\mathbb{E}(Y)$?

Theorem (Mean and variance of the geometric distribution)

If $X \sim G(p)$, then $\mathbb{E}(X) = \dfrac{1}{p}$ and $\mathbb{V}\mathrm{ar}(X) = \dfrac{q}{p^2}$.

Proof left as an exercise: start with the infinite series

$$\frac{1}{1-x} = \sum_{k=0}^{\infty} x^k$$

and use ideas from the proof for the mean and Variance of the Binomial distribution.

Example 21. We have a jar of 20 red and 8 black jelly beans, and we select jelly beans randomly, one at a time, with replacement. Let Y be the number of trials that it takes to get a black jelly bean for the first time.
Find the expected value and the standard deviation of Y.

Solution.

Theorem (Mean and variance of the geometric distribution)

If $X \sim G(p)$, then $\mathbb{E}(X) = \dfrac{1}{p}$ and $\mathbb{V}\mathrm{ar}(X) = \dfrac{q}{p^2}$.

Proof left as an exercise: start with the infinite series

$$\frac{1}{1-x} = \sum_{k=0}^{\infty} x^k$$

and use ideas from the proof for the mean and Variance of the Binomial distribution.

Example 21. We have a jar of 20 red and 8 black jelly beans, and we select jelly beans randomly, one at a time, with replacement. Let Y be the number of trials that it takes to get a black jelly bean for the first time.
Find the expected value and the standard deviation of Y.

Solution.

- *Set up.* The probability of success is $p = \frac{8}{28} = \frac{2}{7}$, so $q = 1 - p = \frac{5}{7}$.

Theorem (Mean and variance of the geometric distribution)

If $X \sim G(p)$, then $\quad \mathbb{E}(X) = \dfrac{1}{p} \quad$ and $\quad \mathbb{V}\mathrm{ar}(X) = \dfrac{q}{p^2}$.

Proof left as an exercise: start with the infinite series

$$\frac{1}{1-x} = \sum_{k=0}^{\infty} x^k$$

and use ideas from the proof for the mean and Variance of the Binomial distribution.

Example 21. We have a jar of 20 red and 8 black jelly beans, and we select jelly beans randomly, one at a time, with replacement. Let Y be the number of trials that it takes to get a black jelly bean for the first time.
Find the expected value and the standard deviation of Y.

Solution.

- *Set up.* The probability of success is $p = \frac{8}{28} = \frac{2}{7}$, so $q = 1 - p = \frac{5}{7}$.
- $\mathbb{E}(X) = \dfrac{7}{2} = 3.5$ *Does this seem right? How could you check?*

Mean and variance of the geometric distribution

Theorem (Mean and variance of the geometric distribution)

If $X \sim G(p)$, then $\quad \mathbb{E}(X) = \dfrac{1}{p} \quad$ and $\quad \mathbb{V}\mathrm{ar}(X) = \dfrac{q}{p^2}$.

Proof left as an exercise: start with the infinite series

$$\frac{1}{1-x} = \sum_{k=0}^{\infty} x^k$$

and use ideas from the proof for the mean and Variance of the Binomial distribution.

Example 21. We have a jar of 20 red and 8 black jelly beans, and we select jelly beans randomly, one at a time, with replacement. Let Y be the number of trials that it takes to get a black jelly bean for the first time.
Find the expected value and the standard deviation of Y.

Solution.

- *Set up.* The probability of success is $p = \frac{8}{28} = \frac{2}{7}$, so $q = 1 - p = \frac{5}{7}$.
- $\mathbb{E}(X) = \dfrac{7}{2} = 3.5$  *Does this seem right? How could you check?*
- $\mathbb{V}\mathrm{ar}(X) = \left(\dfrac{5}{7}\right)\left(\dfrac{7}{2}\right)^2 = \dfrac{35}{4}$ and $\mathrm{SD}(X) = \dfrac{\sqrt{35}}{2} = 2.96$.

Cumulative distribution for a geometrically distributed random variable.

- If $X \sim G(p)$ and $x > 0$ then

$$\mathbb{P}(X > x) = \mathbb{P}(\text{the first } \lfloor x \rfloor \text{ trials result in failures}) = q^{\lfloor x \rfloor} \ ,$$

and so the the cumulative distribution function is

$$F(x) = \mathbb{P}(X \leqslant x) =$$

Cumulative distribution for a geometrically distributed random variable.

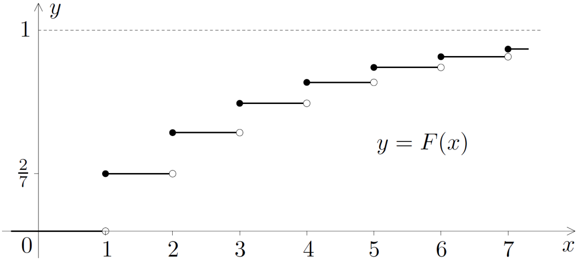
- If $X \sim G(p)$ and $x > 0$ then

$$\mathbb{P}(X > x) = \mathbb{P}(\text{the first } \lfloor x \rfloor \text{ trials result in failures}) = q^{\lfloor x \rfloor},$$

and so the the cumulative distribution function is

$$F(x) = \mathbb{P}(X \leqslant x) = \begin{cases} 1 - q^{\lfloor x \rfloor} & \text{if } x > 0 \\ 0 & \text{if } x \leqslant 0. \end{cases}$$

- Here is a graph of the cumulative distribution function if $X \sim G\left(\frac{2}{7}\right)$.



Does this seem right? Why?

And now, after so much Probability, a teeny tiny taste of Statistics.

The Sign Test

We can use our knowledge of probability to decide whether a comparison between two sets of figures, or a comparison between a set of figures and some standard, can reasonably be put down to chance; or whether, on the other hand, it gives evidence of a genuine difference in the two situations. Here is an example.

Example 22. Suppose that a course has separate algebra and calculus halves. The course authority suspects that students do better on the Algebra half and picks a random sample of students. Their marks are as follows (each column corresponds to a given student).

algebra	7	7	8	3	9	10	8	10	9	7	7	2	9	8	9
calculus	4	7	7	9	10	9	9	7	5	6	4	1	8	8	7

Is there sufficient evidence to say whether students in general do better on the Algebra half?

Example 23, continued.

Solution.

- Set-up: There are two students who scored the same for algebra and calculus: we ignore these. Of the 13 remaining students, 10 did better at algebra and 3 did better at calculus. So (at least on the basis of these figures) we are certainly not going to say that students are better at calculus than algebra.
- We run a Hypothesis test, which is to Stats what a proof by contradiction is to maths.

  What you secretly believe to be true: $H_a =$ 'Students do better at algebra'

  H_0: Suppose that there is actually no real difference in ability: that the class results are purely a matter of chance. Then the number of students who did better at algebra would be a random variable $X \sim B(13, \frac{1}{2})$, and the probability of obtaining results as extreme or more extreme than those observed would be

$$\mathbb{P}(X \geqslant 10) = \left[\binom{13}{10} + \binom{13}{11} + \binom{13}{12} + \binom{13}{13}\right]\left(\frac{1}{2}\right)^{13} = 0.0461 \ .$$

  This is very unlikely; so we do not believe that the difference is pure luck, and we conclude that students are better at algebra than calculus.

For this course we will say that any probability under 5% counts as "very unlikely".